Total Points: [16] [U2HW19 - W29 - Naming Compounds 1]

Q1	A B C D E F
Q2	A B C D E F
Q3	A B C D E F
Q4	A B C D E F
Q5	A B C D E F
Q6	A B C D E F
Q7	A B C D E F
Q8	A B C D E F
Q9	A B C D E F
Q10	A B C D E F
Q11	A B C D E F
Q12	A B C D E F
Q13	A B C D E F
Q14	A B C D E F
Q15	A B C D E F
Q16	A B C D E F

1. If an acid is formed from a polyatomic ion named "carbonate" (CO_3^{2-}), what will be the name of the corresponding acid (H_2CO_3)?

A. Hydrogen carbon trioxide acid

B. Hydrocarbonous acid

C. Percarbonic acid

D. Hydrocarbonic acid

E. Carbonous acid

F. Carbonic acid

2. Many polyatomic ions, such as chlorate (ClO_3^-) and phosphate (PO_4^{3-}), primarily consist of which types of elements bonded together before acquiring an overall charge?

A. A metalloid and a noble gas

B. A metal and a nonmetal element

C. Only hydrogen and a metal

D. Two different metal elements

E. A transition metal and a halogen

F. Multiple nonmetal elements

3. Given that the sulfide ion is S^{2-}, what is the correct chemical name for the compound Fe_2S_3 (Iron can have multiple charges)?

A. Ferrous(II) sulfite

B. Iron(II) trisulfide

C. Iron sulfide

D. Iron(III) sulfide

E. Diiron trisulfide

F. Iron(II) sulfide

4. The acid $HClO_2$ contains the chlorite ion (ClO_2^-). According to the rules for naming polyatomic acids (oxyacids), what is the correct chemical name for $HClO_2$?

A. Perchloric acid

B. Hypochloric acid

C. Hydrochlorous acid

D. Hydrochloric acid

E. Chlorous acid

F. Chloric acid

5. A compound is named 'Lead (IV) oxide'. What does the '(IV)' indicate about the lead in this specific compound?

A. Lead is in the fourth period of the periodic table.

B. There are four oxygen atoms bonded to lead.

C. The lead ion has a +4 charge.

D. The oxide ion has a -4 charge.

E. It is the fourth most common isotope of lead.

F. There are four lead atoms in the formula.

6. An atom or group of atoms that has gained electrons, resulting in a negative charge, is known as a(n) ______, and is typically formed from a ______ element:

A. polyatomic ion, metalloid

B. anion, nonmetal

C. cation, metal

D. cation, nonmetal

E. isotope, noble gas

F. anion, metal

7. What is the correct chemical name for the simple ionic compound K_2O, formed from potassium and oxygen?

A. Potassium(I) oxide

B. Potassic oxide

C. Dipotassium monoxide

D. Potassium peroxide

E. Potassium(II) oxide

F. Potassium oxide

8. A compound with the formula PCl_3 consists of phosphorus and chlorine (both nonmetals). Based on its composition, how would this compound primarily be classified and named?

A. As an oxyacid, because it contains a nonmetal and chlorine.

B. As a simple binary ionic compound, with chlorine named "chlorate".

C. As a covalent compound, using numerical prefixes.

D. As a polyatomic ion, requiring a Roman numeral for phosphorus.

E. As a binary acid, using the "hydro-" prefix.

F. As an ionic compound, naming the cation and then the anion ending in "-ide".

9. In the systematic naming of binary covalent compounds like SO_2, which rule applies to the use of Greek prefixes?

A. Prefixes are only used if both elements have more than one atom.

B. Roman numerals are used instead of prefixes for the first element.

C. 'Mono-' is omitted for the first element, but always used for the second if there is only one atom.

D. 'Mono-' is always used for the first element if there is only one atom.

E. The prefix 'uni-' is used instead of 'mono-' for the first element.

F. Prefixes are only used for the second element, never the first.

10. When naming an ionic compound composed of a metal and a polyatomic ion, such as Na_3PO_4 (where PO_4^{3-} is phosphate), how is the polyatomic ion part named?

A. Its name is used as is, without alteration.

B. A 'hydro-' prefix is added to its name.

C. Its ending is changed to -ic if it originally ended in -ate.

D. Its ending is changed to -ide.

E. Its ending is changed to -ous if it originally ended in -ite.

F. Numerical prefixes (di-, tri-) are added based on its implied count in the formula.

11. What is the correct chemical name for the compound $Al(NO_3)_3$ (NO_3^- is the nitrate ion).?

A. Aluminum nitrite

B. Aluminum nitrate

C. Aluminum(III) nitrate

D. Aluminic trioxide nitride

E. Nitrogen aluminate

F. Aluminum trinitrate

12. The systematic name for a binary acid, such as H_2Se (hydroselenic acid), always includes which of the following combinations of prefix and suffix for the nonmetal part?

A. No prefix, -ous acid

B. hypo- prefix, -ite acid

C. No prefix, -ate acid

D. hydro- prefix, -ic acid

E. per- prefix, -ic acid

F. hydro- prefix, -ous acid

13. Consider the acid H_3PO_4 (containing the phosphate ion, PO_4^{3-}) and the acid HI. Which statement accurately describes a key difference in their naming conventions?

A. Both are named by simply stating "hydrogen" followed by the anion name (e.g., hydrogen phosphate).

B. Both are named using the "hydro-" prefix because they start with hydrogen.

C. H_3PO_4 (from phosphate) is named phosphoric acid (no "hydro-"), while HI is named hydroiodic acid.

D. H_3PO_4 is named using the "hydro-" prefix, while HI is not.

E. H_3PO_4 is named perphosphoric acid, and HI is named hypoiodous acid.

F. H_3PO_4 is named phosphorous acid, and HI is named iodic acid.

14. In a simple ionic compound formed between a metal like Barium (Ba) and a nonmetal like Fluorine (F), what is the correct ending for the anion's name (fluorine part)?

A. -ide

B. -ate

C. -ous

D. -ic

E. -ite

F. -ine

15. A key characteristic used to initially identify a compound as a potential acid from its chemical formula is the presence of:

A. Hydrogen as the first element.

B. A metal as the first element.

C. Oxygen as the last element.

D. Carbon as the central atom.

E. A halogen as the first element.

F. A polyatomic ion as the only component.

16. What is the correct chemical name for the binary covalent compound N_2O_4?

A. Nitrogen(IV) oxide

B. Dinittric tetroxide

C. Dinitrogen oxide

D. Nitrogen oxide

E. Dinitrogen tetroxide

F. Nitrogen tetraoxygen